An informational theory of privacy

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Outline

1. Introduction
2. Model
3. Results
4. Applications
5. Conclusion

Privacy is hot topic

Privacy in news and life:

- government surveillance
- business surveillance
  - e-business: data as side product of any transaction
  - loyalty cards
- voluntary provision of personal data (facebook, twitter, mobile phone)
- micro targeting in election campaigns

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What is at stake? trade-offs? mechanisms?

What is privacy (in this paper)?

- ability to take actions without being observed, and having interactions with others confined to the intended recipients
- (many other definitions)
- privacy $\subset$ asymmetric information

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Is there an economic rational for privacy?

Economics of privacy: Chicago school (Stigler 80, Posner 81)

- privacy = asymmetric information
- asymmetric information = inefficiency (Akerlof, Mirrlees etc.)

$\Rightarrow$ privacy = inefficiency

Popular debate: "Nothing to hide"

- reiterated by Google, facebook, NSA etc.
  - If you have nothing to hide, you do not need privacy.
  - If you have something to hide, you should not do it.
- (similar to Chicago school)

This paper

- our model:
  - information is not perfect (correlation: statistical discrimination as in Phelps 72, Arrow 73)
  - privacy affects behavior
- main result: privacy can be efficient even when considering *informational effects only*
- other results:
  - lack of privacy changes behavior in one direction ("chilling effects")
  - effectiveness of privacy intrusion is easily overstated
  - privacy is redistributive (affects different people differently)
  - mandating privacy might be necessary to avoid unraveling

A simple story

- Alice thinks drugs should be legalized and wants to write on her facebook profile about that.
- She is also looking for a job.
- employers do not want to hire drug users
- positive correlation between stand on legalization (θ_i) and drug use (τ_i)
- Two cases: The employer can see what Alice does online, or not.

Model I

- n individuals
- individual i has type (θ_i, τ_i)
  - $\theta_i \sim \mathcal{N}(0, 1)$
  - τ_i positively correlated with θ_i
    (formally: distribution of τ_i at θ_i' fosd distribution at θ_i''$<\theta$')

Model II

1. information aggregation stage:
   - i observes (θ_i, τ_i)
   - i chooses $p_i \in \{0, 1\}$
   - policy $p \in \{0, 1\}$ is implemented with probability m/n where m is number of individuals choosing $p_i = 1$
   - payoff for i: $p\theta_i$
2. interaction stage
   - opposing player (OP) chooses action A (aggressive) or M (mild) against i
   - payoff OP: A gives τ_i, M gives 0
   - payoff i: A gives $-\delta(\tau_i)$, M gives 0
   - δ weakly increasing

no privacy: OP knows p_i
privacy: OP has only prior

Results: Chilling effect

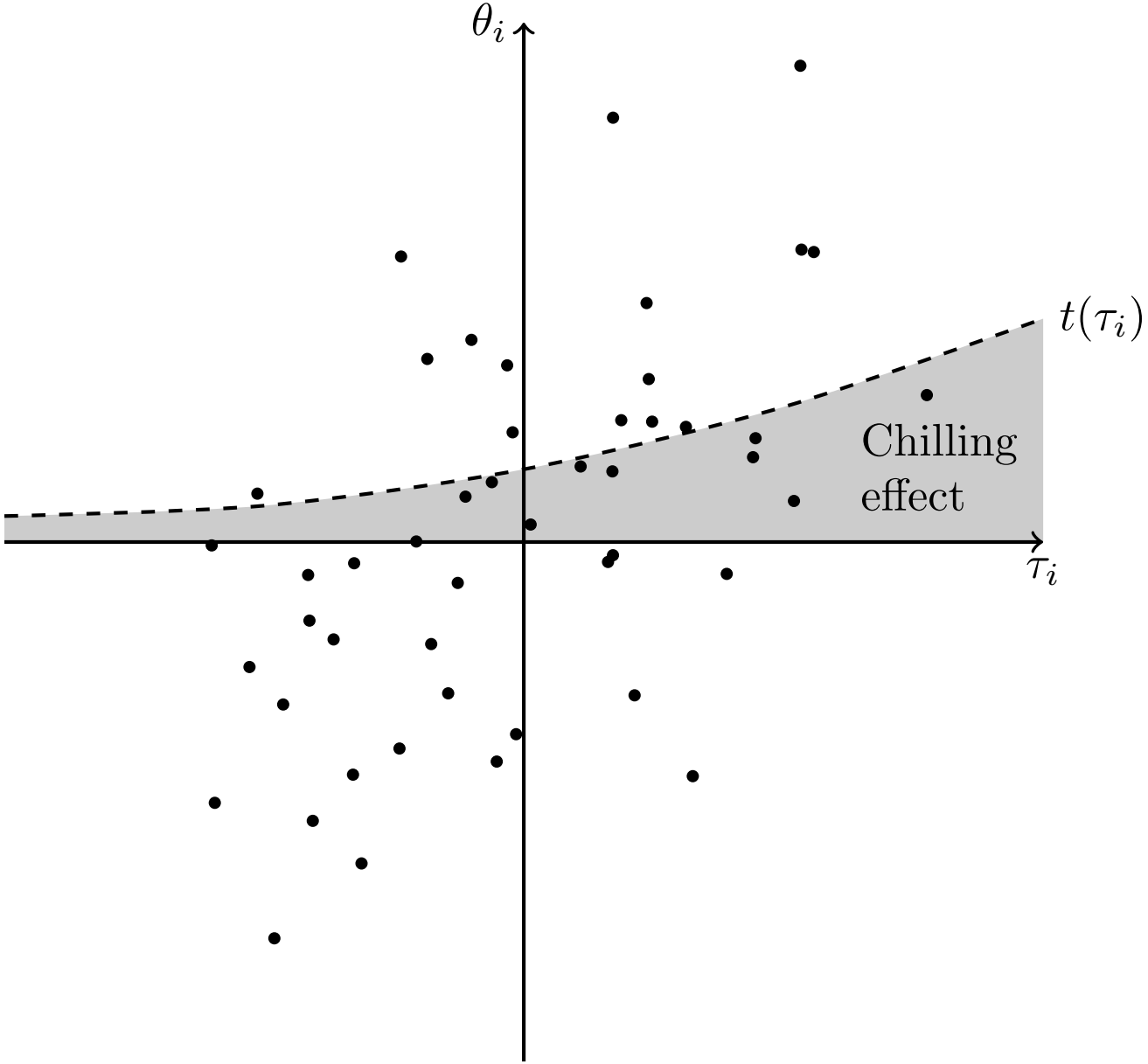
- individuals use cutoff strategies: $p_i = 1$ iff $\theta_i \geq t(\tau_i)$

Proposition (Chilling effect)

With privacy $t^p(\tau_i) = 0$.
Without privacy $t^{np}(\tau_i) \geq 0$ and $t^{np}(\tau_i)$ is increasing.

- without privacy individuals with $\theta_i \in [0, t(\tau_i))$ change their behavior ("are chilled")
- without privacy OP plays M against $p_i = 0$ and A against $p_i = 1$
  (if privacy affects OP behavior and pure strategy equilibrium)

Equilibrium without privacy



Welfare results

Lemma (Consumer surplus)

Expected consumer surplus in the information aggregation stage is maximal at the privacy cutoff $t^p(\tau_i) = 0$.

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- welfare includes OP payoff (OP might loose from privacy)

Theorem (Welfare)

Let $\delta' > 0$.
For n sufficiently large, expected consumer surplus is higher under privacy and OP payoff is the same under privacy and no privacy.
Let the disutility of A be $-r\delta(\tau_i)$. For r sufficiently high, consumer surplus is higher under privacy and OP payoff is the same under privacy and no privacy.

Privacy is redistributive

- individuals with $\theta_i < 0$ loose from privacy
- individuals that are chilled $\theta_i \in [0, t^{np}(\tau_i)$ gain a bit from privacy
- individuals with $\theta_i > t^{np}(\tau_i)$ gain double from privacy (decision, OP interaction)

Surveillance performs worse than expected

- seems as eliminating privacy could give OP huge benefit
- but: behavior change (chilling) might reduce correlation between p_i and τ_i
- technical assumption (TA): $\mathbb{E}[\tau_i | \theta_i = 0] \geq 0$
  (if OP knew θ_i and $\theta_i \geq 0$, A would be best response)

Lemma

Assume TA.
OP's payoff without privacy is lower if individuals use t^{np} than if they used $t^p = 0$.

Surveillance performs worse than expected

Hello [redacted],

We thought you might be interested in knowing that customers who bought "200g*0.01g Mini Digital Pocket Scale for Jewelry Kitchen Gram Oz Ct" also bought these items.

Customers Also Bought...

100 Plastic Resealable Grip Seal Bags...

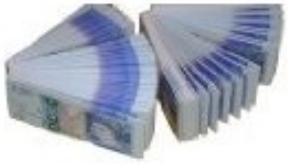
Sold by: AGOODBUYFROMME

100 Grip Seal Bags 2.25 x 3 Inch...

Sold by: Express Goods UK

250 Raw Filter TIPS card booklets...

Sold by: Martins Deals



S AND S Â£20 Pound Note Design...

Sold by: BARGAIN BASE

100 Grip Seal Bags 1.5" x 2.5"

Sold by: Swoosh Supplies

JUICY Juicy Jays Kingsize...

Sold by: Premier Life Store

S AND S 750 Roaches Roach Filter...

Sold by: BargainShop_London

Correlation between τ_i and θ_i

- assume pure strategy equilibrium without privacy
- take convex combination of correlated and uncorrelated distribution

$$\Gamma^{\lambda}_{\theta_i}(\tau_i) = \lambda \Gamma_{\theta_i}(\tau_i) + (1-\lambda)\bar{\Gamma}(\tau_i) \qquad \lambda \in [0,1].$$

$$\bar{\Gamma}(\tau_i) = \int_{\Re} \Gamma_{\theta_i}(\tau_i)\, d\Phi(\theta_i).$$

Proposition

There exist $0 < \underline{\lambda} < \bar{\lambda} < 1$ such that

- for $\lambda \leq \underline{\lambda}$ privacy and no privacy are welfare equivalent and
- for $\lambda \in (\underline{\lambda}, \bar{\lambda}]$ privacy leads to strictly higher welfare than no privacy. The equilibrium for $\lambda = \bar{\lambda}$ is in pure strategies.

More stuff in the paper I

- Extension: privacy as opt in
  - individuals can choose whether to keep p_i private or not
  - multiple equilibria
  - privacy is not a robust equilibrium: unraveling
- Extension: endogenous information aggregation mechanism q
  - q is chosen to maximize consumer surplus taking (no) privacy into account
- Extension: defensive action
  - Alice can hire a lawyer to help her with the employer
  - i can take costly defensive action that reduces the disutility of A and lowers the payoff of OP (regardless of his action)
  - equilibria where OP is strictly better off with privacy

More stuff in the paper II

Alternative setups for stage 1:

- private decision
  - stage 1: no information aggregation, i.e. no externalities from p_j on i
  - additional result: privacy is welfare optimal if correlation between θ_i and τ_i is not too high
- state matching
  - stage 1: payoff of each i is $p\theta$ where θ is an unknown state and each i has a noisy signal θ_i
  - same results but privacy makes every i better off as chilling inhibits efficient information aggregation

Discussion: When is privacy bad?

- biased information aggregation q: chilling can counteract bias
- externalities:
  - insider trading
  - video surveillance to deter vandalism might prevent other crimes

Application: Chilling effects in election polls

- secret ballot: election is private and influences decision
- poll: less private (interviewer), less influential
  $\rightarrow$ chilling effects
- Bischoping and Schuman (1992)

Application: Credit scoring

repayment probability (τ_i) is not directly observable

- Consider two characteristics (θ_i) that are predictive of τ_i: education and music taste
- Low education and a preference for rap music predict low repayment probability.
- There is a chilling effect in both cases, but in the first we might consider it desirable!?
- Should the bank be allowed to use data on music taste? ("Equal Credit Opportunity Act" outlaws "redlining" in the US)
- blacklist vs. whitelist

Application: Working in committees

- committee is debating two policies (e.g. raise interest rates or not)
- debate and vote can either be in secret or in public
- correlation between policy preference θ_i and competence τ_i
- members worry about being perceived as incompetent; advocate less radical positions
- Fed is forced to publish minutes of FOMC meetings since 1993;
  increase in conformity and a decrease of disagreement with the chairman (Meade and Stasavage 2008)
- Thomas Hoenig, President of the Kansas City Fed: "The tape has had some chilling effect on our discussions. I see a lot more people reading their statements."

Conclusion

- we give a simple model to understand privacy from the perspective of information economics
  - no privacy leads to chilling effects
  - privacy can be welfare optimal
  - privacy is redistributive (similar to freedom of speech, Friedman 72)
- the model leads to interesting policy questions
  - blacklist vs. whitelist
  - how open should government be?
  - mandate vs. option of privacy
  - which kind of data should be accessible by who?
- we identify crucial elements to address these questions (correlation, behavior change, threat potential)

Literature

- Chicago school (Stigler 80, Posner 81)
- statistical discrimination (Phelps 72, Arrow 73)
- Hirshleifer effect (Hirshleifer 71)
- intrinsic motivation + image concerns and privacy (Daughety and Reinganum 2010 , Ali and Benabou *working paper*)
- equilibrium effects of privacy (Cummings et al. *working paper*)
- many papers in specific IO settings
  - dynamic pricing if purchase history is known/unknown
    - sophisticated vs. naive consumers
  - differentiated good competition with known/unknown location